A UNIVERSALLY INJECTIVE ETALE MORPHISM OF ALGEBRAIC SPACES IS AN OPEN IMMERSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement begins with a slogan block (a chapter-internal display environment): it is part of the counted statement span, verbatim, and its definition is covered by the reference/slogan/history preamble span, so the delivered text typesets the statement without the slogan line.

Editorial note. The six references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Stability of etaleness and universal injectivity under base change. Lemmas lemma-base-change-etale and lemma-base-change-universally-injective, used at proof lines 10958-10962. Reduces the claim to the case where Y is a scheme.
- The diagonal is etale, a monomorphism and surjective for a universally injective morphism. Lemma lemma-universally-injective, used at proof lines 10963-10965. Supplies the three properties of the diagonal.
- A surjective flat locally finitely presented monomorphism is an isomorphism. Spaces, Lemma spaces-lemma-surjective-flat-locally-finite-presentation, used at proof lines 10966-10968. Concludes that the diagonal is an isomorphism, i.e. X is separated over Y.
- A separated locally quasi-finite morphism with representable target has representable source. Proposition proposition-locally-quasi-finite-separated-over-scheme, used at proof lines 10969-10971. Makes X a scheme, so that the scheme theorem applies.
- The fundamental theorem for etale morphisms of schemes: etale radicial equals open immersion. Etale Morphisms, Theorem etale-theorem-etale-radicial-open, used at proof lines 10972-10974. Concludes that f is an open immersion.

Source reference. Original theorem 025G: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2813–2813. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0466: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5461–5461. Navigation pointer into the Stacks

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Source reference. Original lemma 03MW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4764–4764. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 040X: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5272–5272. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original proposition 03XX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5164–5164. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05VM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7556–7556. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be an étale and universally injective morphism of algebraic spaces over S. Then f is an open immersion.*

Proof. Let $T \to Y$ be a morphism from a scheme into Y. If we can show that $X \times_Y T \to T$ is an open immersion, then we are done. Since being étale and being universally injective are properties of morphisms stable under base change (see Lemmas 0466 and 03MW) we may assume that Y is a scheme. Note that the diagonal $\Delta_{X/Y} : X \to X \times_Y X$ is étale, a monomorphism, and surjective by Lemma 040X. Hence we see that $\Delta_{X/Y}$ is an isomorphism (see Spaces, Lemma 05VM), in particular we see that X is separated over Y. It follows that X is a scheme too, by Proposition 03XX. Finally, $X \to Y$ is an open immersion by the fundamental theorem for étale morphisms of schemes, see Étale Morphisms, Theorem 025G. □